Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**3**
Electrons (Ne)	**3**
Hopping (t)	**1.0**
Interaction (U)	**4.75**
n-body order	**3**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-1.11969618**
Energy E1	**-1.11969618**
Gap (E1-E0)	**0.000000**
<S^2>	**0.750000**
S	**0.5000**
<Sz>	**0.250000**
Multiplicity	**2.0**
Spin label	**Doublet (S=1/2)**

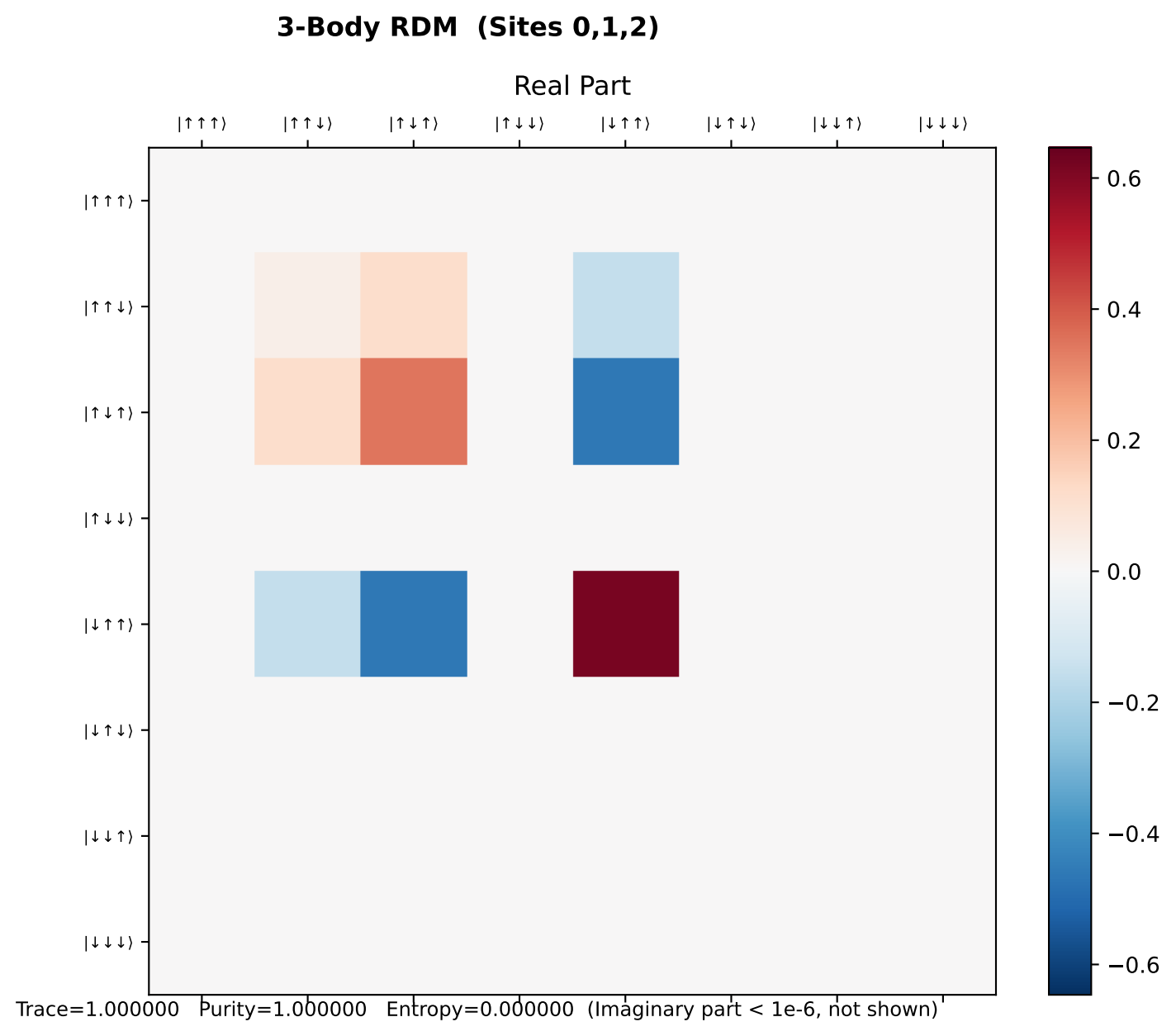
Double occupancy

<D> total	**0.185102**
<D> / site	**0.061701**

RDM and GME statistics for this run (3-body clusters)

Hilbert space dim : 2^3 = 8
Number of clusters : 3
Pure clusters : 3
Mixed clusters : 0
Clusters with GME (>0) : 3

Purity gamma = Tr(rho^2) : mean = 1.000000 [min 1.000000 , max 1.000000]
Von Neumann entropy : mean = 0.000000 [min 0.000000 , max 0.000000]
Rank of rho : mean = 1.00 [min 1 , max 1]
C_GME (best estimate) : mean = 0.385137 [min 0.385137 , max 0.385137]


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	1.0000000000
Von Neumann entropy	0.0000000000
Eigenvalue min	-7.976722e-17
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(0,)	(1, 2)	0.52668453
(0, 1)	(2,)	0.92583480
(0, 2)	(1,)	0.54748067

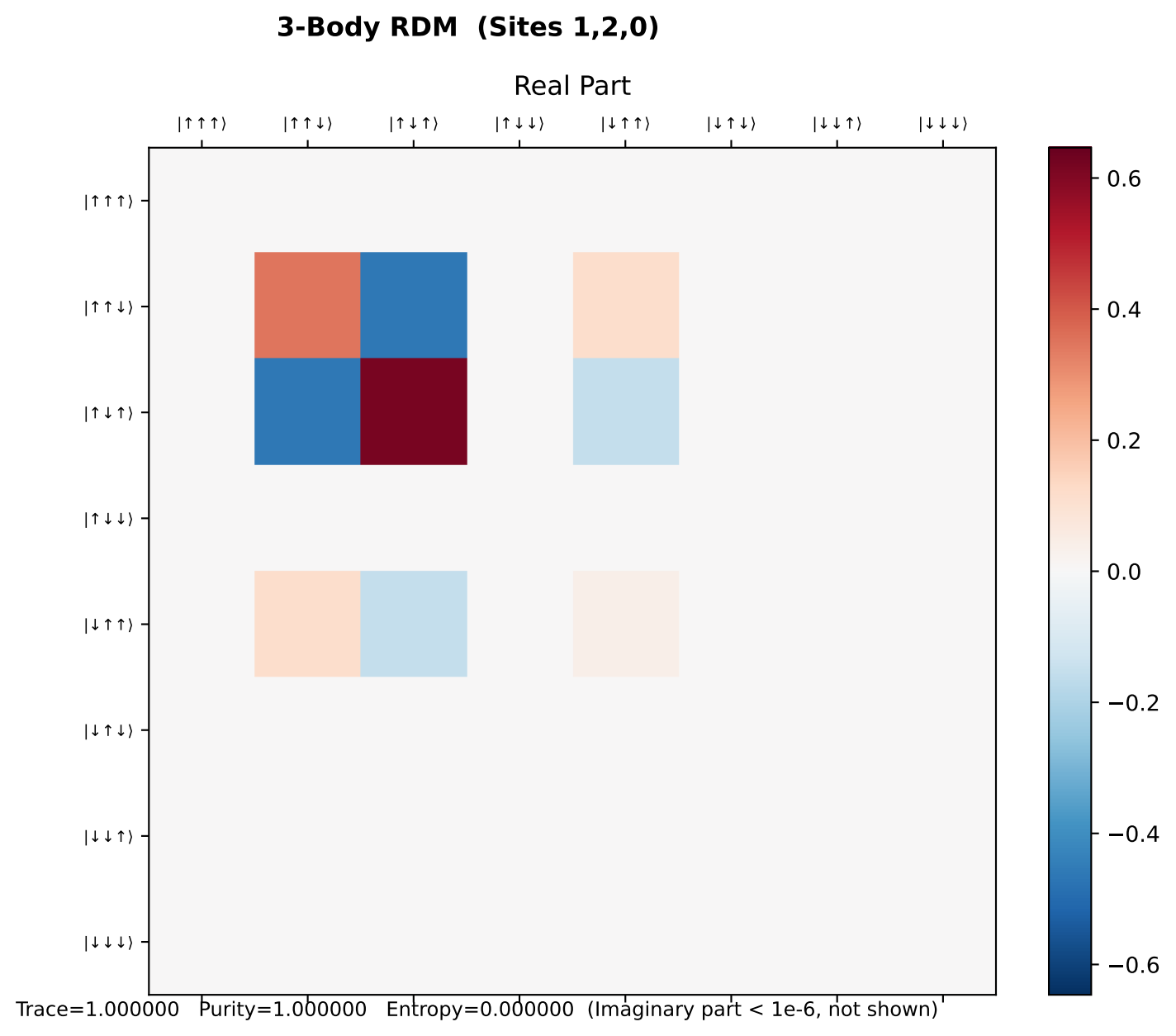
STATUS: GME EXISTS

C_GME = 0.38513685
GME confirmed across ALL cuts.

Convex Roof Analysis
==========

Lower Bounds:
  C_LB   = 0.00000000
C_GME (unitary optimization):
  C_stiefel (unitary opt) = 0.38513685   BEST
C_GME (unitary optimization - Expoenetial Map):
Upper Bound:
  C_UB (eigendecomp) = 0.38513685

Status: GME


3-Body RDM (Sites 1,2,0)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,0)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.717102e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (1,)         | (2, 0)       | 0.92583480
  (1, 2)       | (0,)         | 0.54748067
  (1, 0)       | (2,)         | 0.52668453

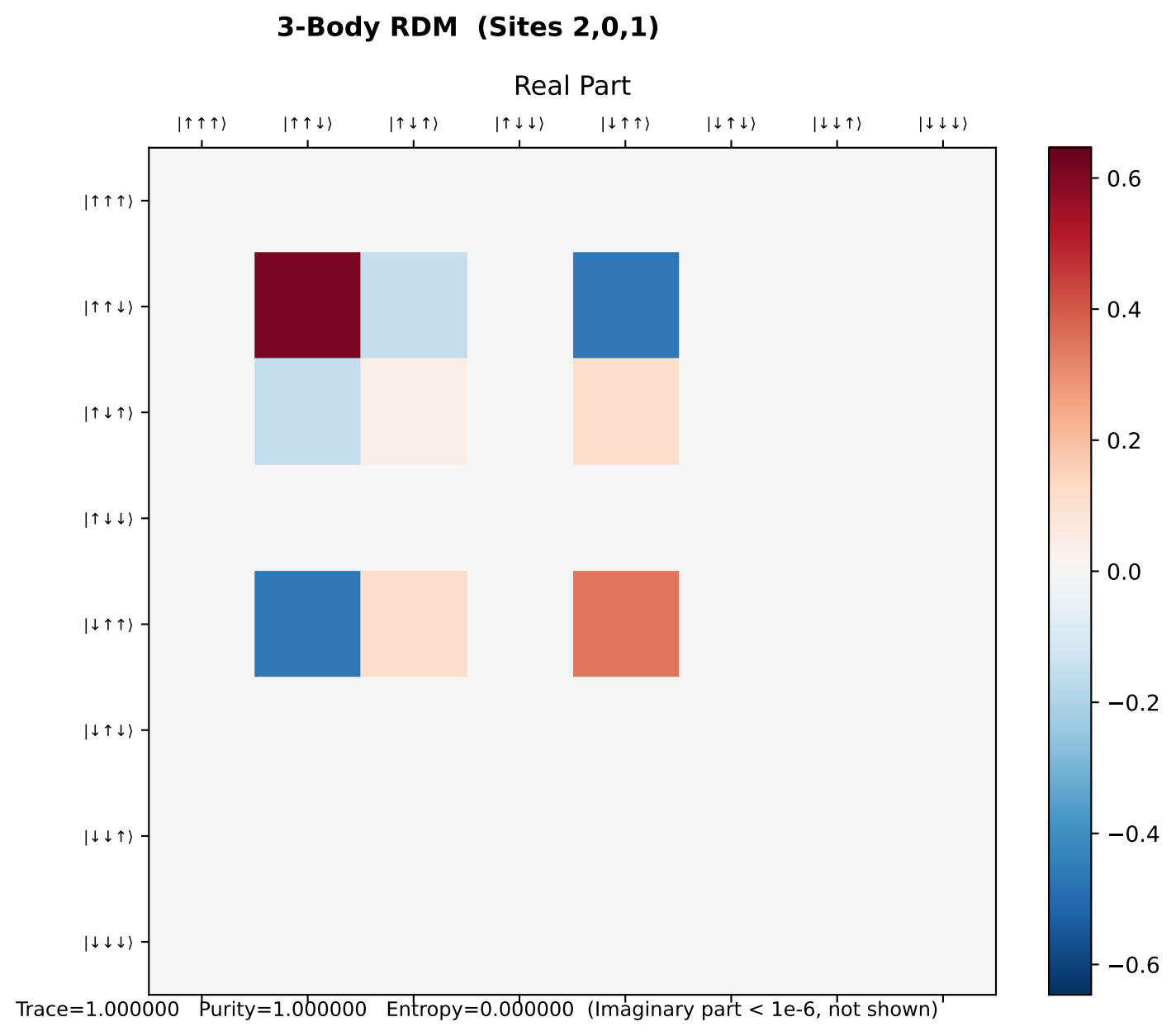
STATUS: GME EXISTS
------------------------------------

C_GME = 0.38513685
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.38513685   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.38513685

  Status: GME
```


3-Body RDM (Sites 2,0,1)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,0,1)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -8.734196e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (0, 1)       | 0.54748067
  (2, 0)       | (1,)         | 0.52668453
  (2, 1)       | (0,)         | 0.92583480

STATUS: GME EXISTS
------------------------------------

C_GME = 0.38513685
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

  Lower Bounds:
    C_LB    = 0.00000000
  C_GME (unitary optimization ):
    C_stiefel (unitary opt) = 0.38513685   BEST
  C_GME (unitary optimization - Expoenetial Map):
  Upper Bound:
    C_UB (eigendecomp)  = 0.38513685

  Status: GME
```